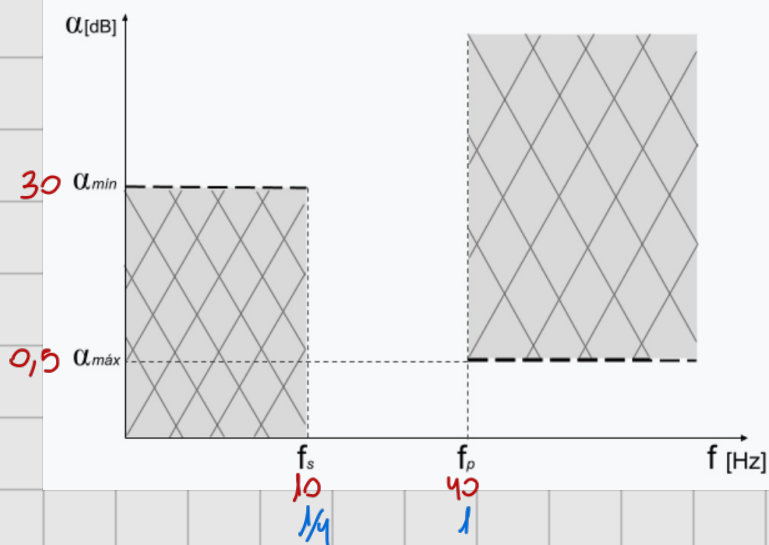
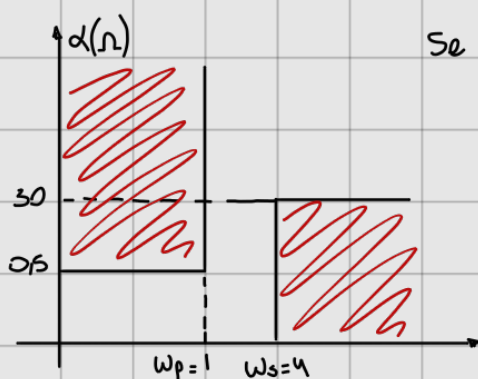




$\alpha_{\text{Max}} [\text{dB}]$	$\alpha_{\text{min}} [\text{dB}]$	$f_p [\text{kHz}]$	$f_s [\text{kHz}]$
0.5	30	40	10

1) Filtro pas alto.

Plantilla de transformación



Se ha aplicado una normalización de frecuencia
 $N_{\Omega} = (2\pi)(40 \text{ kHz})$

Calculamos los parámetros del filtro Chebyshev.

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 C_m^2(\omega)}$$

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \epsilon^2 \cosh^2(N \cdot \alpha \cosh(\omega))}}$$

Hallamos los parámetros:

Parámetros de ripple.

$$\epsilon^2 = 10^{\frac{\alpha_{\text{max}} - \alpha_{\text{min}}}{20}} - 1 = 10^{\frac{0.5 - 30}{20}} - 1 = 0.122 //$$

$$\alpha_n = 10 \cdot \log(1 + \epsilon^2 \cosh^2(n \cdot \cosh^{-1}(\omega_s))) = 10 \log(1 + (0.122) \cosh^2(n \cosh^{-1}(4)))$$

$$\epsilon^2 = 0.122 \text{ y } n = 3$$

Utilizo la relación recursiva de los polinomios de Chebyshev. $n=1: \alpha_n = 4, + \otimes$
 $n=2: \alpha_n = 20, + \otimes$
 $n=3: \alpha_n = 38,6 \text{ (✓)} > \alpha_{\text{min}}$

$$C_n(\omega) = 2\omega C_{n-1}(\omega) - C_{n-2}(\omega) \text{ con } C_0(\omega) = 1 \text{ y } C_1(\omega) = \omega$$

$$\text{Para } n=2: C_2(\omega) = 2\omega C_1(\omega) - C_0(\omega) = 2\omega(\omega) - 1 = 2\omega^2 - 1 //$$

$$\text{Para } n=3: C_3(\omega) = 2\omega C_2(\omega) - C_1(\omega) = 2\omega(2\omega^2 - 1) - \omega = 4\omega^3 - 2\omega - \omega = 4\omega^3 - 3\omega //$$

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$H(j\omega)H(-j\omega) = \frac{1}{1 + \epsilon^2 C_3(\omega)} = \frac{1}{1 + \epsilon^2 (4\omega^3 - 3\omega)^2} = \frac{1}{1 + \epsilon^2 (16\omega^6 - 24\omega^4 + 9\omega^2)} =$$

$$H(j\omega)H(-j\omega) = \frac{1/\epsilon^2}{16\omega^6 - 24\omega^4 + 9\omega^2 + 1/\epsilon^2} =$$

$$H(s)H(-s) = \frac{1/\epsilon^2}{-s^6(16) - s^4(24) - s^2(9) + 1/\epsilon^2}$$

$H(s)$ tiene un denominador cuyo polinomio es de grado 3 y $H(-s)$ también.

$$H(s)H(-s) = \frac{1/16\epsilon^2}{-s^6 - s^4(24/16) - s^2(9/16) + 1/16\epsilon^2} =$$

$$H(s)H(-s) = \frac{1}{(s^3 + as^2 + bs + c)(-s^3 + as^2 - bs + c)}$$

$$H(s)H(-s) = \frac{1}{-s^6 - bs^4 + a^2s^4 + acs^2 - bs^4 - b^2s^2 + cas^2 + c^2}$$

$$H(s)H(-s) = \frac{1}{-s^6 + (a^2 - 2b)s^4 + (2ac - b^2)s^2 + c} = \frac{1/16\epsilon^2}{-s^6 - s^4(3/2) - s^2(9/16) + 1/16\epsilon^2}$$

$$\frac{1}{16\epsilon^2} = c^2 \rightarrow c = 0,715 //$$

$$-9/16 = (2ac - b^2) \rightarrow b^2 = 2ac + 9/16 \rightarrow \left(\frac{a^2 + 3/2}{2}\right)^2 = 2ac + 9/16$$

$$-3/2 = (a^2 - 2b) \rightarrow \frac{a^2 + 3/2}{2} = b \quad a^4 + 3a^2 + 9/4 = 8ac + 9/4$$

$$a^4 + 3a^2 - a(5,72) = 0$$

$$a^3 + 3a - 5,72 = 0 \rightarrow a = 1,25 //$$

$$b = 1,53 //$$

Resultado

$$H(s) = \frac{1}{s^3 + as^2 + bs + c}$$

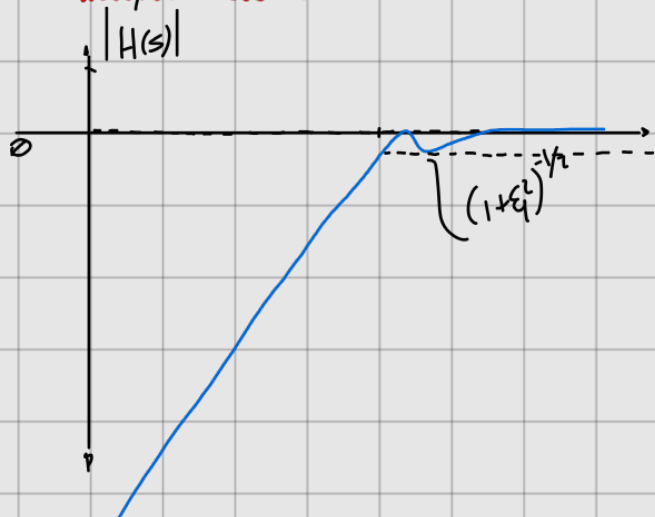
$$H_L(s) = \frac{0,715}{s^3 + (1,25)s^2 + (1,535)s + 0,715}$$

Transformo el circuito a un paso más, $K(s) = 1/s$

$$H_{HP}(s) = \frac{0,715}{1/5 + 1,25 \cdot 1/5^2 + 1,53 \cdot 1/5 + 0,715} = \frac{s^3(0,715)}{s^3(0,715) + s^2(1,53) + s(1,25) + 1}$$

$$H_{HP}(s) = \frac{s^3}{s^3 + s^2(2,14) + s(1,75) + 1,4}$$

2) Respuesta de módulo



Respuesta de fase

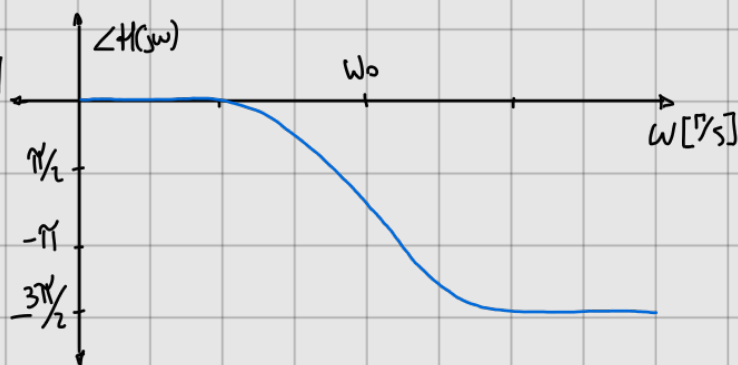
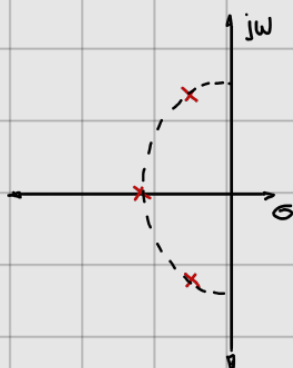


Diagrama de polos y ceros.

$$H_{LP}(s) = \frac{0,715}{s^3 + (1,75)s^2 + (1,535)s + 0,715}$$



$$H_{HP}(s) = \frac{s^3}{s^3 + s^2(2,14) + s(1,75) + 1,4}$$



los polos del Chebyshev No tienen todos el mismo módulo, están sobre una elipse.

3) primero factorizo la transferencia

$$H_{HP}(s) = \frac{s^3}{s^3 + s^2(2,14) + s(1,75) + 1,4} = \frac{s^3}{(s + \alpha)(s^2 + sb + c)}$$

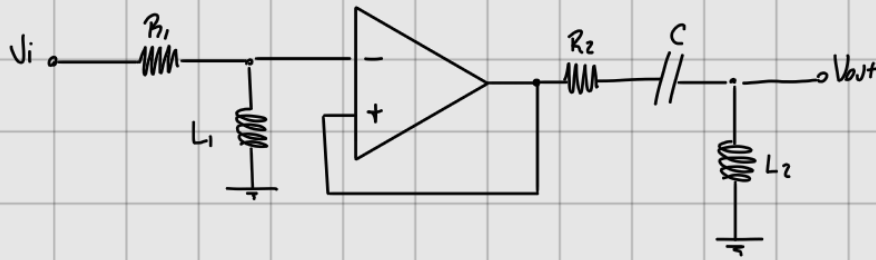
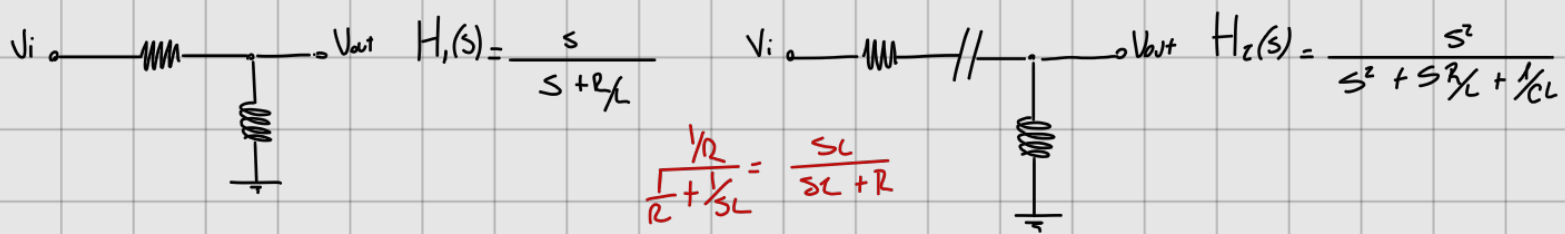
para $s = -1,6$

$$(-1,6)^3 + (-1,6)^2(2,14) + (-1,6)(1,75) + 1,4 \approx 0 \quad \alpha = -1,6$$

-1,6	1	2,14	1,75	1,4
	-1,6	-0,864	-1,416	
	1	0,54	0,886	-0,016

$$H_{HP}(s) = \frac{s^3}{(s - 1,6)(s^2 + s(0,54) + 0,886)}$$

$$H_{HP}(s) = \frac{s}{(s + 1,6)} \cdot \frac{s^2}{s^2 + s(0,54) + (0,886)}$$



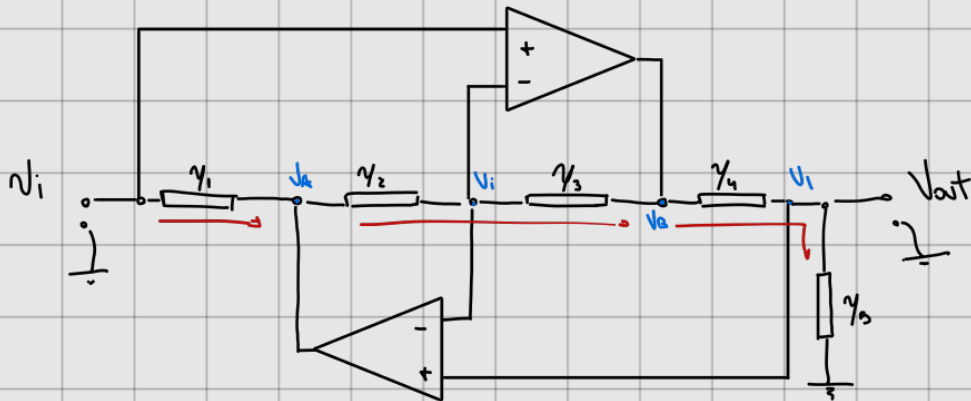
$$H(s) = \frac{s}{s + R_1/L_1} \cdot \frac{s^2}{s^2 + sR_2/L_2 + 1/CL_2}$$

$$\frac{R_1}{L_1} = 1,6, \quad \frac{R_2}{L_2} = 0,54, \quad \frac{1}{CL_2} = 0,886$$

$$R_1 = 1, \quad R_2 = 1 \rightarrow \frac{1}{L_1} = 1,6, \quad \frac{1}{L_2} = 0,54, \quad \frac{1}{C_2} = \frac{0,886}{0,54}$$

4)

GIC Antoniou



Impedancia de entrada

$$* I_{in} = (V_i - V_A) \gamma_1, \quad * V_A = V_B \cdot \frac{\gamma_4}{\gamma_4 + \gamma_5} \rightarrow V_B = V_A \cdot \frac{\gamma_4 + \gamma_5}{\gamma_4}$$

$$\frac{V_i(\gamma_1) - I_{in}}{\gamma_1} = V_A$$

$$* V_i(\gamma_2 + \gamma_3) = V_A \gamma_2 + V_B \gamma_3$$

$$V_i(\gamma_2 + \gamma_3) = \frac{V_i(\gamma_1 \gamma_4) - I_{in} \gamma_2}{\gamma_1} + V_i(\gamma_4 + \gamma_5) \left(\frac{\gamma_4}{\gamma_4} \right)$$

$$\cancel{V_i(\gamma_2 + \gamma_3)} - \cancel{V_i \gamma_2} - \cancel{V_i \gamma_3} - V_i \frac{\gamma_3 \gamma_5}{\gamma_4} = I_{in} \frac{\gamma_2}{\gamma_1}$$

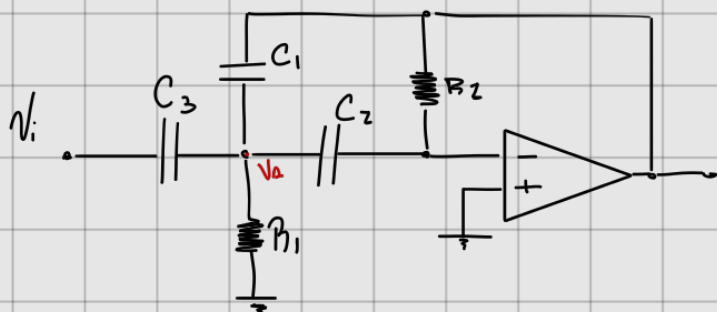
$$I_{in} = V_i \cdot \frac{\gamma_1 \gamma_3 \gamma_5}{\gamma_2 \gamma_4}$$

$$Z_{in} = \frac{V_i}{I_{in}} \rightarrow Z_{in} = \frac{\gamma_1 \gamma_4}{\gamma_3 \gamma_5}$$

$$Z_{in} = \frac{Z_1 Z_2 Z_3}{Z_4 Z_5}$$

Para que tenga comportamiento inductivo, $Z_4 = \frac{1}{sC}$ entonces nos queda $Z_{in} = \frac{Z_1 Z_2 Z_3 (sC)}{Z_2} = sL_{eq}$

5) HFB (Passivos)



$$V_A(sC_1 + sC_2 + sC_3 + \frac{1}{R_1}) = V_i(sC_3) + V_o(sC_1)$$

$$V_o(sC_2) = -V_o \frac{1}{R_2}$$

$$V_A = -V_o \frac{1}{sC_2 R_2}$$

$$-V_o \left(\frac{s(C_1 + C_2 + C_3)R_1 + 1}{sC_2 R_1 R_2} \right) = V_i(sC_3) + V_o(sC_1)$$

$$-V_o \left(\frac{s(C_1 + C_2 + C_3)R_1 + 1 + s^2 C_1 C_2 R_1 R_2}{sC_2 R_1 R_2} \right) = V_i(sC_3)$$

$$|H(s)| = \frac{s^2 C_2 C_3 R_1 R_2}{s^2 C_1 C_2 R_1 R_2 + s(C_1 + C_2 + C_3)R_1 + 1}$$

$$|H(s)| = \frac{C_3}{C_1} \cdot \frac{s^2}{s^2 + s \frac{(C_1 + C_2 + C_3)}{C_1 C_2 R_2} + \frac{1}{C_1 C_2 R_1 R_2}}$$

No Normalizado

$$C_1 = C_2 = C_3 // \frac{C_1 + C_2 + C_3}{C_1 C_2 R_2} = 0,54$$

$$k = \frac{C_3}{C_1} // \omega_0^2 = \frac{1}{C_1 C_2 R_1 R_2} // \frac{\omega_0}{Q} = \frac{C_1 + C_2 + C_3}{C_1 C_2 R_2}$$

$$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_1 R_2}} // Q = \frac{\sqrt{C_1 C_2 R_1 R_2}}{R_1 (C_1 + C_2 + C_3)}$$

$$\frac{3R}{C^2 R_2} = 0,54 // C = \frac{3}{(0,54)R_2} // \frac{1}{C_1 C_2 R_1 R_2} = 0,886$$

$$\frac{(0,54)^2}{3^2 (0,886)} = \frac{R_1}{R_2} // \frac{1}{3^2 \frac{1}{R_1 R_2} (0,54)^2} = 0,886$$

$$R_2 = 1$$

Análisis sensibilidad

$$C_1 = \frac{1}{\omega_0^2 C_2 R_1 R_2}$$

$$S_{\omega_0}^{C_1} = \frac{\omega_0}{C_1} \frac{\partial(C_1)}{\partial(\omega_0)} = \frac{\omega_0}{C_1} \frac{1}{\partial(\omega_0)} \left(\frac{1}{C_2 R_1 R_2} \omega_0^{-2} \right) = (-2) \frac{\omega_0}{C_1} \cdot \frac{1}{\omega_0^2 C_2 R_1 R_2} = \frac{(-2)}{\omega_0^2 C_1 C_2 R_1 R_2} = (-2) \frac{\omega_0}{\omega_0^2}$$

$$S_{\omega_0}^{C_1} = -2 \text{ como es } < 0 \text{ } \omega_0 \text{ es en sentido contrario. } \frac{\Delta C_1}{C_1} = (-2) \cdot \frac{\Delta \omega_0}{\omega_0}$$

$$R_1 = \frac{\sqrt{C_1 C_2 R_2}}{q(C_1 + C_2 + C_3)} \quad S_q^{R_1} = \frac{q}{R_1} \cdot \frac{\partial R_1}{\partial q} = \frac{q}{R_1} \cdot \frac{1}{q} \left(\frac{\sqrt{C_1 C_2 R_2}}{q(C_1 + C_2 + C_3)} \right) = \frac{1}{R_1} \cdot \frac{\sqrt{C_1 C_2 R_2}}{q(C_1 + C_2 + C_3)}$$

$$S_q^{R_1} = -2$$

$$R_2 = \frac{R_1 q^2 (C_1 + C_2 + C_3)^2}{C_1 C_2}$$

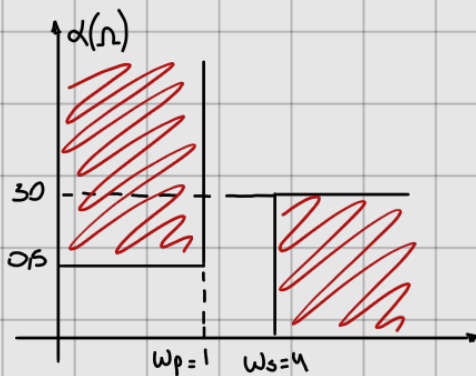
$$S_q^{R_2} = \frac{q}{R_2} \frac{\partial R_2}{\partial q} = \frac{q}{R_2} \cdot (2q) \cdot \frac{R_1 (C_1 + C_2 + C_3)^2}{C_1 C_2} = \frac{(2)}{R_2} \cdot \frac{R_1 q^2 (C_1 + C_2 + C_3)^2}{C_1 C_2} = 2 \cdot \frac{R_2}{R_2}$$

$$S_q^{R_2} = 2$$

Bonus

Red del mismo orden con una aproximación por máxima planicidad.

Plantilla de transformación



máxima planicidad

$\epsilon_1^2 = 10^{\frac{\alpha_{\max}}{10}} - 1 = 0,122$			
$\alpha_n = 10 \log(1 + \epsilon_1^2 w_s^{2n}) = 10 \log(1 + (0,122)(4)^{2n})$			
n			
2	15	$< \alpha_{\max}$	✗
3	26,99	$< \alpha_{\max}$	✗
4	39,02	$> \alpha_{\max}$	✓

Como es evidente, para poder implementar este filtro por máxima planicidad, habría que implementar un filtro de un orden mayor al del caso Chebyshev.